

Bayesian Neural Network - CE23B004

The objective of this project is to develop an advanced ground motion model using a **Bayesian Neural Network (BNN)** to predict seismic intensity measures (IMs) for subduction zone earthquakes. Unlike traditional deterministic models, this BNN framework provides a good treatment of uncertainty by decomposing it into aleatoric and epistemic components.

Dataset and Preprocessing

The pipeline processes the NGA-Subduction Mega-Flatfile, which initially contains 71,340 earthquake records. The model identifies 105 distinct spectral acceleration periods ranging from $T = 0.010$ s to $T = 10.0$ s.

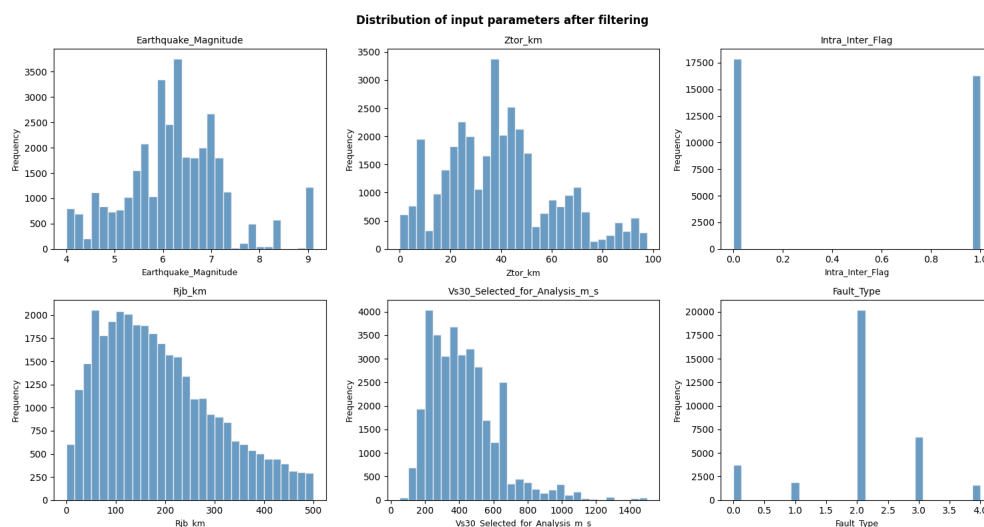
The data is categorized into primary inputs and target outputs:

- **Primary Input Features:** These include Earthquake Magnitude M_w , Depth to top of rupture Z_{tor} , Intra/Inter-slab flag, Joyner-Boore distance R_{jb} , V_{s30} , and Fault Type.
- **Target Output Variables:** The pipeline targets Peak Ground Acceleration (PGA), Peak Ground Velocity (PGV), and the 105 identified Spectral Acceleration $[Sa(T)]$ periods.

Data Filtering:

- **Seismological Constraints:** Records are limited to magnitudes $M_w > 4$, rupture depths $0 < Z_{tor} < 100$ km, and distances $0 < R_{jb} < 500$ km.
- **Site and Event Classification:** Only records with Intra/Inter-slab flags of 0 or 1 are retained to exclude ambiguous events, and site conditions are limited to $V_{s30} < 1500$ m/s.
- **Intensity Measure Validation:** To prevent errors during log-transformation, the pipeline removes any records where PGA, PGV, or any of the 105 spectral acceleration values are non-positive or exceed 5g.

After applying these scalar and spectral filters, 37,117 records are removed, leaving a final clean dataset of **34,223 records**.



Feature Engineering

To better capture the non-linear physics of ground motion, five engineered features are added to the input vector:

- **Magnitude Scaling:** A Magnitude_Squared (M_w^2) term is calculated to account for non-linear magnitude scaling.
- **Log-Transformations:** Natural log transforms are applied to distance (Log_Rjb), site conditions (Log_Vs30), and depth (Log_Ztor) with a +1 offset to handle surface ruptures).
- **Interactions:** A Magnitude-Distance interaction term (M_w times $\ln(R_{jb})$) is engineered to model the dependence of distance attenuation on earthquake size.

This results in a total of **11 refined input features**.

All final input features:

Feature	Type	Description	Why It Matters
Earthquake Magnitude (Mw)	Raw	Moment magnitude of the earthquake	Bigger quakes = stronger shaking
Ztor (km)	Raw	Depth to top of the rupture zone	Shallow ruptures typically cause stronger shaking at surface
Intra/Inter Flag	Raw	Type of subduction event: intraplate (0) or interface (1)	These two event types behave very differently seismically
Rjb (km)	Raw	Horizontal distance from site to fault projection	Shaking decays with distance — key scaling factor
Vs30 (m/s)	Raw	Average shear-wave velocity of top 30 m of soil	Soft soils amplify shaking; rock does not
Fault Type	Raw	Type of fault mechanism	Affects radiation pattern of seismic waves
Magnitude Squared	Engineered	M_w^2	Captures the nonlinear (curved) relationship between magnitude and intensity
Log(Rjb)	Engineered	Natural log of distance	Distance attenuation follows a log-linear trend
Log(Ztor)	Engineered	Log of rupture depth	Linearises the depth scaling
Log(Vs30)	Engineered	Log of site velocity	Site amplification is approximately log-linear
$M_w \times \text{Log(Rjb)}$	Engineered	Interaction term	Captures that large magnitude quakes attenuate differently over distance

Model Architecture

The model implements a **Bayesian Neural Network** designed to handle noise in seismic data.

The architecture features a significantly increased capacity over previous versions:

- **Shared Trunk:** 3 hidden layers with 256 neurons each using ELU activation.
- **Heteroscedastic Heads:** The shared trunk branches into two separate heads:
 1. **Mean Head:** Predicts the 107 mean values μ for the intensity measures.
 2. **Log-Variance Head:** Predicts 107 log-variances $\log \sigma^2(\text{aleatory})$, allowing the model to learn scenario-specific irreducible noise.
- **Total Parameters:** Approximately 379,308 trainable parameters.

Component	Detail
Input layer	11 features (standardised to zero mean, unit variance)
Hidden trunk	3 Bayesian layers, each with 256 neurons and ELU activation — shared across both output heads
Mean head	1 Bayesian layer → 107 mean predictions (μ)
Log-variance head	1 Bayesian layer → 107 log-variance predictions ($\log \sigma^2 \text{ aleat}$)
Total parameters	~190,000

Uncertainty Quantification

The BNN framework allows for a formal decomposition of uncertainty:

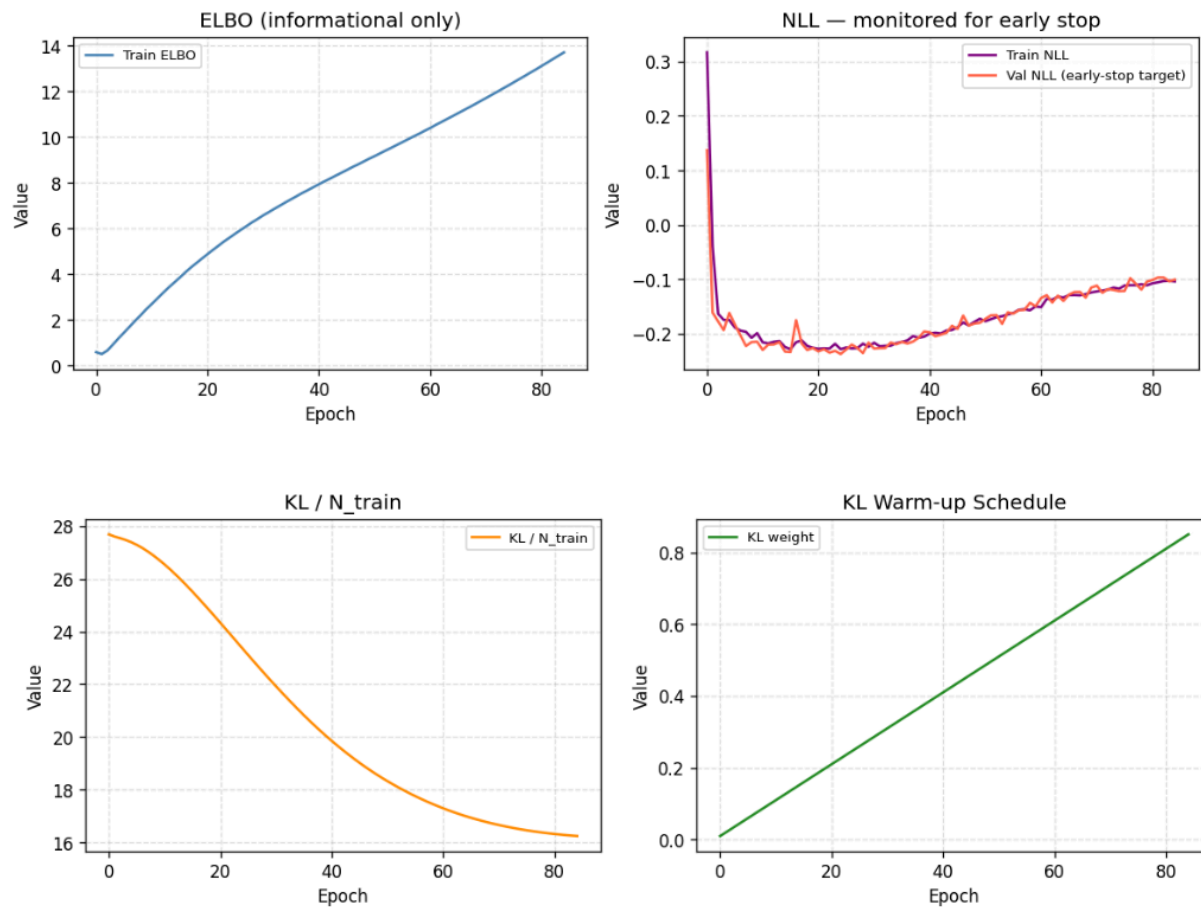
- **Aleatoric (σ_{aleat}):** Captured by the log-variance head, representing irreducible data noise.
- **Epistemic (σ_{epis}):** Derived from the variance of predictions across multiple stochastic forward passes (Monte Carlo sampling), representing uncertainty due to model weight distributions.
- **Total Uncertainty:** Calculated as $\sigma_{\text{total}} = \sqrt{\sigma_{\text{epis}}^2 + (\sigma_{\text{aleat}}^2)}$.

Training Methodology

The data is split into three subsets to ensure robust evaluation:

- **Training (70%):** 23,956 records.
- **Validation (15%):** 5,133 records.
- **Testing (15%):** 5,134 records.
- **Optimizer:** Adam optimizer with an initial learning rate of 0.001.
- **Loss Function:** Evidence Lower Bound (ELBO), combining Negative Log-Likelihood (NLL) and KL-divergence with a 100-epoch warmup period for the KL weight.
- **Early Stopping:** Monitor validation NLL to ensure generalization.

Model Training



Four Plots shows how training goes over time.

ELBO: Rises linearly as the KL warm-up ramp increases the KL weight each epoch, this is expected and not a sign of divergence. The ELBO is intentionally not monitored for early stopping here.

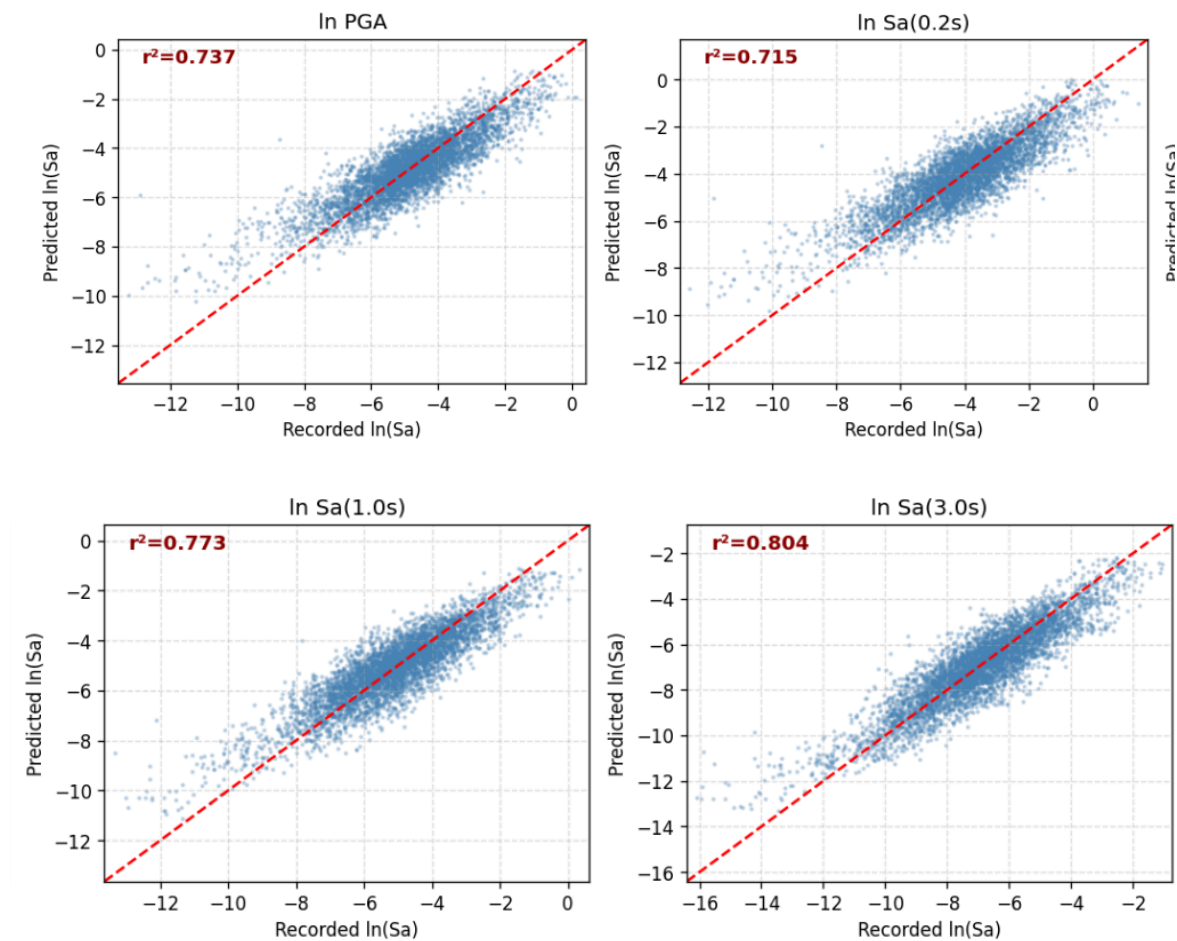
NLL: Drops sharply from ~ 0.31 at epoch 1 to ~ -0.20 by epoch ~ 15 , then slowly rises back toward -0.10 before the model stops at epoch ~ 85 . The initial spike is the warm-up phase; the subsequent rise suggests mild overfitting or the growing KL penalty beginning to constrain the data-fit. Train and val NLL track each other closely throughout, indicating no serious overfitting.

KL / N_train: Decreases steadily from ~ 28 to ~ 16 over training. This means the posterior weight distributions are collapsing toward the prior as the model finds a balance between data fit and regularisation where the model retains only the uncertainty it needs.

KL Warm-up Schedule: Linear ramp from $0 \rightarrow \sim 0.85$ over 85 epochs, confirming early stopping fired before the warmup completed its full 100-epoch schedule. The model was saved at the best NLL point, which occurred mid-warmup, suggesting the optimal trade-off between data fit and KL regularisation was found before full KL weight was applied.

Metric	PGA	Sa(0.2s)	Sa(1.0s)	Sa(3.0s)	Overall
R	0.8586	0.8456	0.8794	0.8969	0.9327
RMSE	0.8244	0.9030	0.8090	0.8649	0.8558
MAE	0.6418	0.7034	0.6329	0.6760	0.6645
PP	0.2376	0.2174	0.2744	0.3112	0.4088

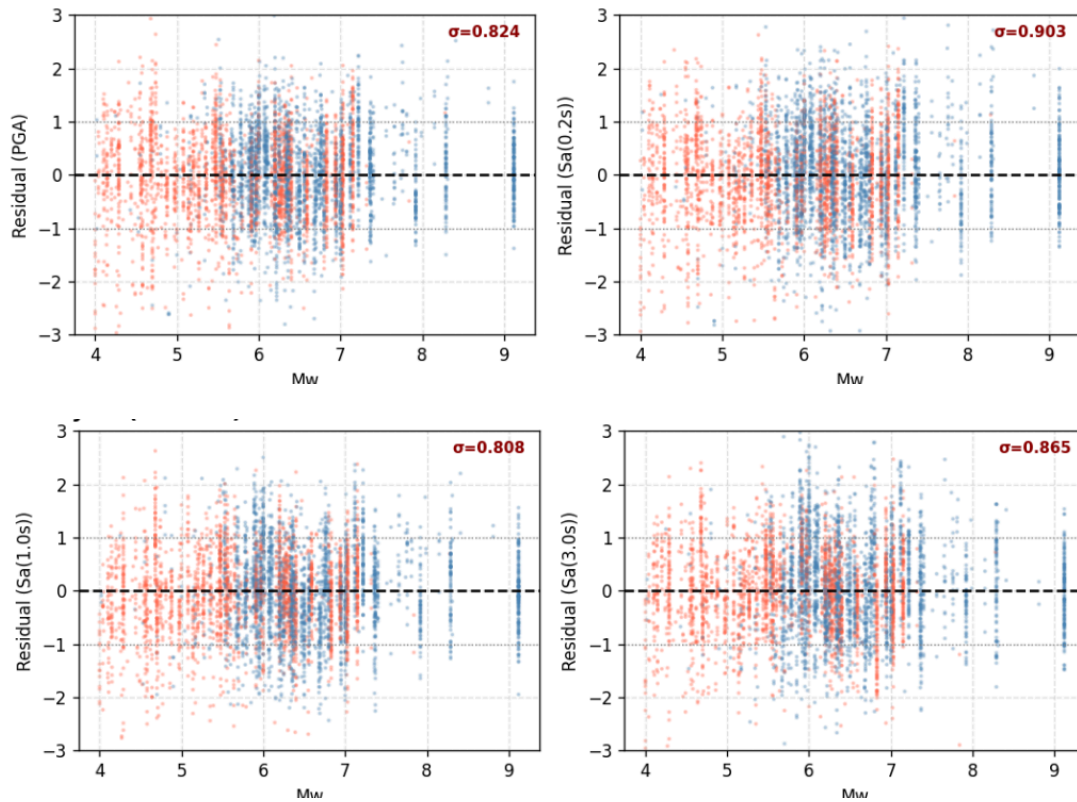
Observed vs Predicted



All four plots show points clustering reasonably well around the 1:1 line with no systematic bias, confirming the model is well-calibrated. r^2 improves consistently from 0.715 at Sa(0.2s) to 0.804 at Sa(3.0s), reflecting the well-known pattern that long-period ordinates are more predictable from source and path parameters. Short-period predictions (PGA, Sa(0.2s)) show wider scatter due to higher sensitivity to site effects and high-frequency stochasticity. The symmetric spread across the full amplitude range indicates the BNN captures both weak and strong motion scenarios without amplitude-dependent bias.

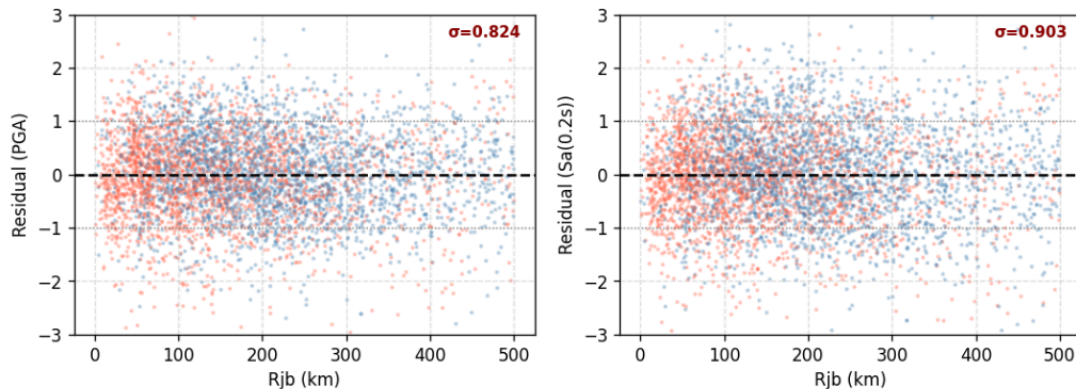
Residual Analysis

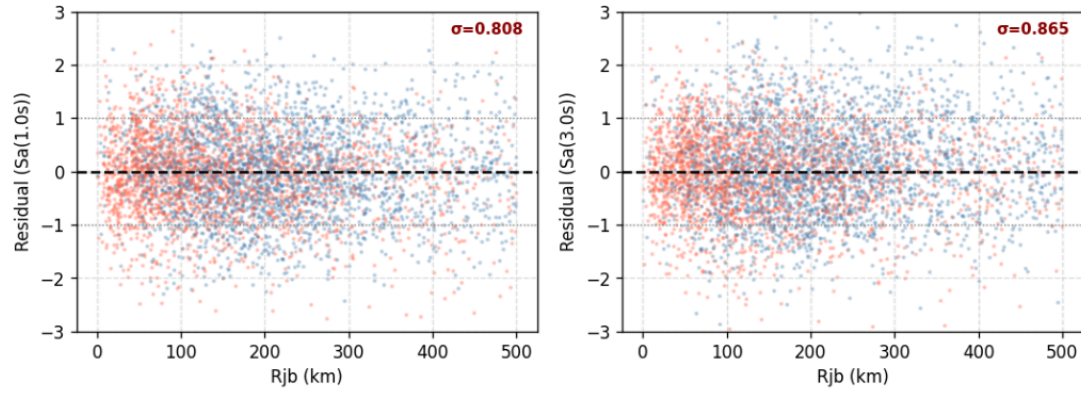
1. wrt. Magnitude



Residuals are centered near zero across the full magnitude range (4–9.1) with no visible trend, confirming the model captures magnitude scaling without systematic over- or under-prediction. Scatter is higher at moderate magnitudes (M_w 5–7) due to greater data density, while the sparse high-magnitude records ($M_w > 8$) show slightly elevated spread — expected given limited training data in that range. σ is consistent across all four periods (~ 0.81 – 0.90).

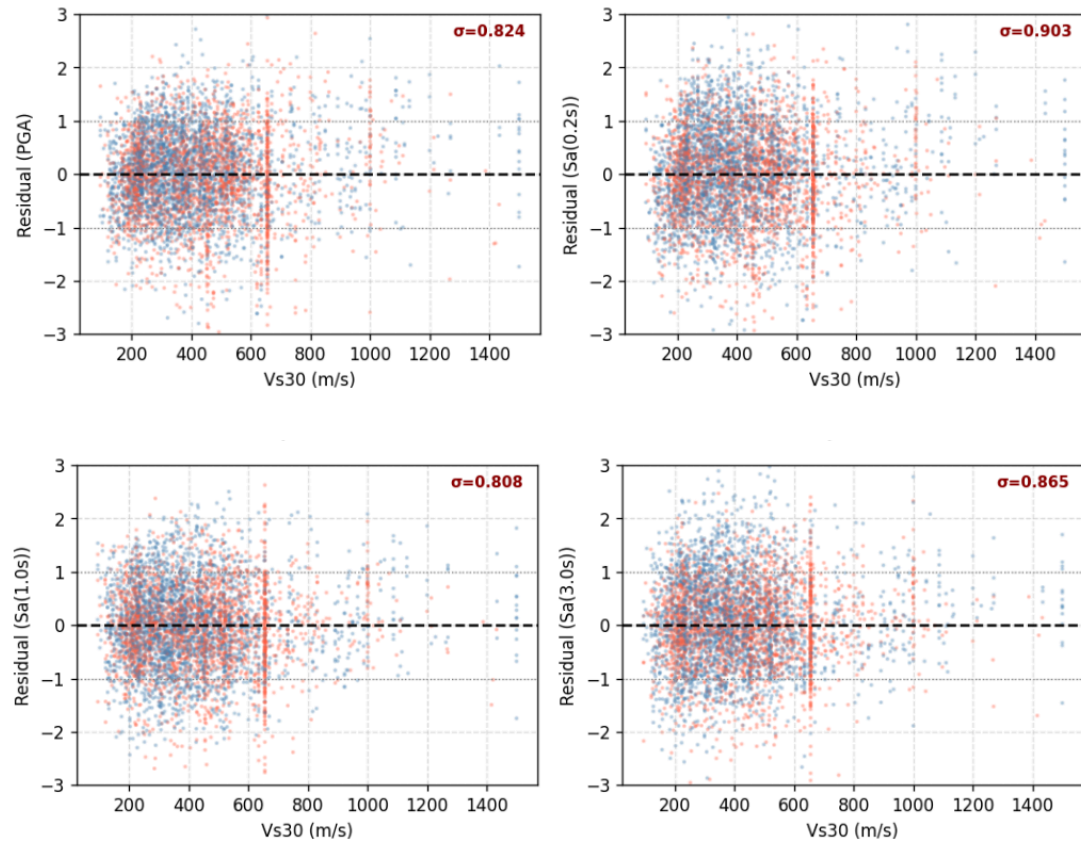
2. wrt. Rjb distance





Mean residuals remain near zero across the full distance range (0–500 km), indicating the model correctly captures ground motion attenuation with distance. There is a modest increase in scatter at larger distances (>300 km), likely due to sparser records and greater path variability at long ranges. No distance-dependent bias is evident, which validates the log-distance engineering features included in the model.

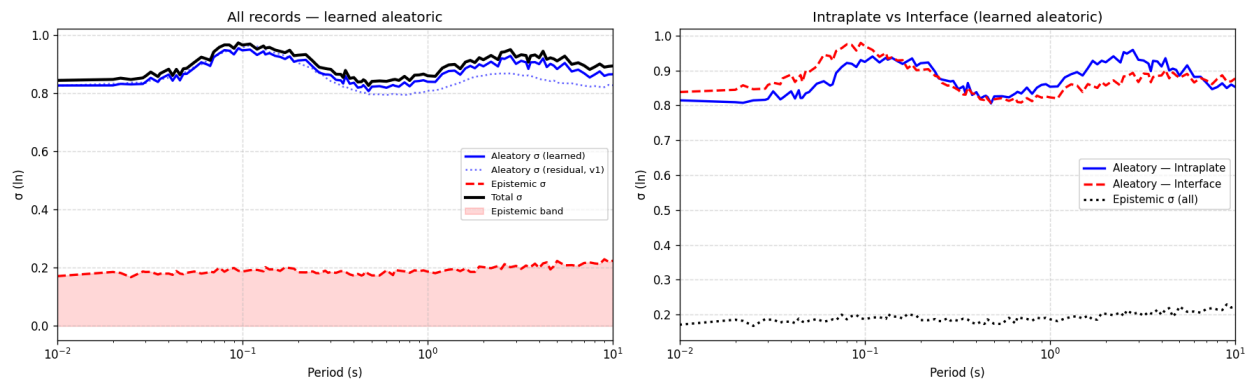
3. Wrt. Rjb distance



Residuals are unbiased across the soft-to-rock site spectrum (180–1500 m/s), demonstrating the model adequately represents site amplification effects. A notable vertical cluster of interface (red) points near Vs30 ≈ 600 m/s reflects the concentration of recording stations at that site class rather than any model deficiency.

Uncertainty Decomposition

In a heteroscedastic BNN, total uncertainty is decomposed into two distinct components. **Aleatory uncertainty** (σ_{aleat}) represents irreducible natural variability in ground motion — the inherent randomness of the earthquake process that cannot be reduced by collecting more data. It is learned directly by the network's log-variance output head, making it scenario-dependent rather than a fixed post-hoc residual. **Epistemic uncertainty** (σ_{epis}) reflects model uncertainty — the uncertainty in the network weights themselves, quantified as the standard deviation of mean predictions across 100 stochastic forward passes. This component can theoretically be reduced with more training data. The total uncertainty combines both as $\sigma_{\text{total}} = \sqrt{(\sigma_{\text{epis}}^2 + \sigma_{\text{aleat}}^2)}$, following the standard additive variance decomposition.

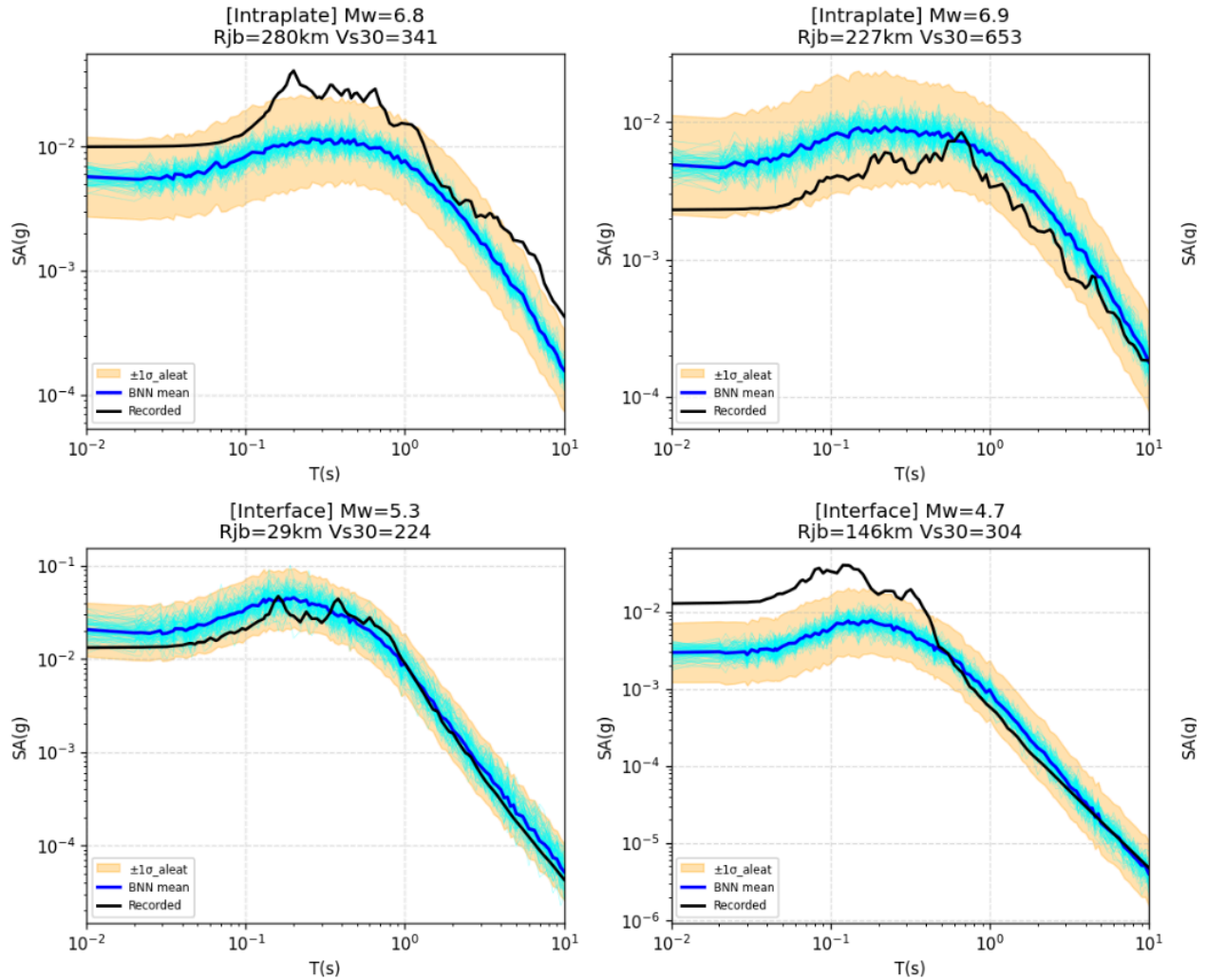


Left panel : The learned aleatory σ (solid blue, ~ 0.82 – 0.97 ln) is consistently higher than the residual-based v1 estimate (dotted blue, ~ 0.80 – 0.85 ln), confirming the heteroscedastic head correctly assigns higher uncertainty to scenarios that are genuinely harder to predict rather than averaging it globally. The epistemic σ (dashed red, ~ 0.18 -- 0.22 ln) remains well below aleatory throughout, which is physically expected. The total σ (black) closely tracks aleatory, confirming epistemic contributions modestly but meaningfully. Both aleatory and total σ peak around $T = 0.1$ -- 0.2 s, corresponding to the resonant frequency range where site amplification effects are most variable and hardest to predict from Vs30 alone. There is a slight dip at intermediate periods (~ 0.5 – 1.0 s) before rising again at long periods (>3 s), reflecting the increased long-period variability due to large-magnitude subduction records in the dataset.

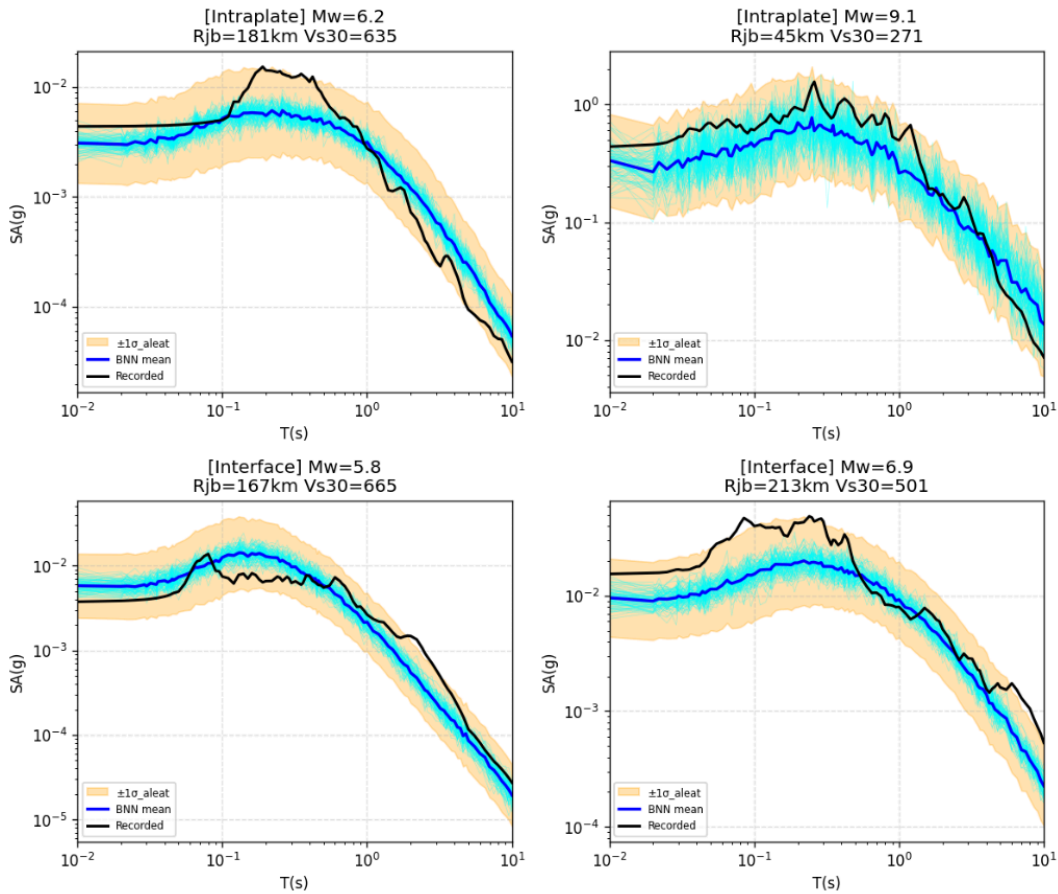
Right panel — Intraplate vs Interface: Interface events (red dashed) consistently show higher aleatory σ than intraplate events (solid blue) across most periods, particularly peaking at ~ 0.1 s ($\sigma \approx 0.97$ vs ~ 0.91). This is physically meaningful; interface earthquakes involve larger, more complex rupture surfaces with greater variability in directivity, slip distribution, and wave propagation through the subducting slab, making their ground motions inherently more variable. The epistemic σ (dotted black) is period-smooth and well below both aleatory curves, confirming the model uncertainty is properly controlled and does not drive the total.

Ensemble Prediction Plots

Each panel shows 100 stochastic forward passes through the BNN (cyan lines), capturing epistemic uncertainty from weight distributions. The orange shading represents $\pm 1\sigma_{\text{aleat}}$ — the learned heteroscedastic aleatoric band — which varies by scenario and period. The BNN mean (blue) is the average of all 100 passes, while the black line is the actual recorded spectrum. Together these three layers fully decompose prediction uncertainty: cyan spread = epistemic, orange band = aleatory, gap between blue and black = prediction error.

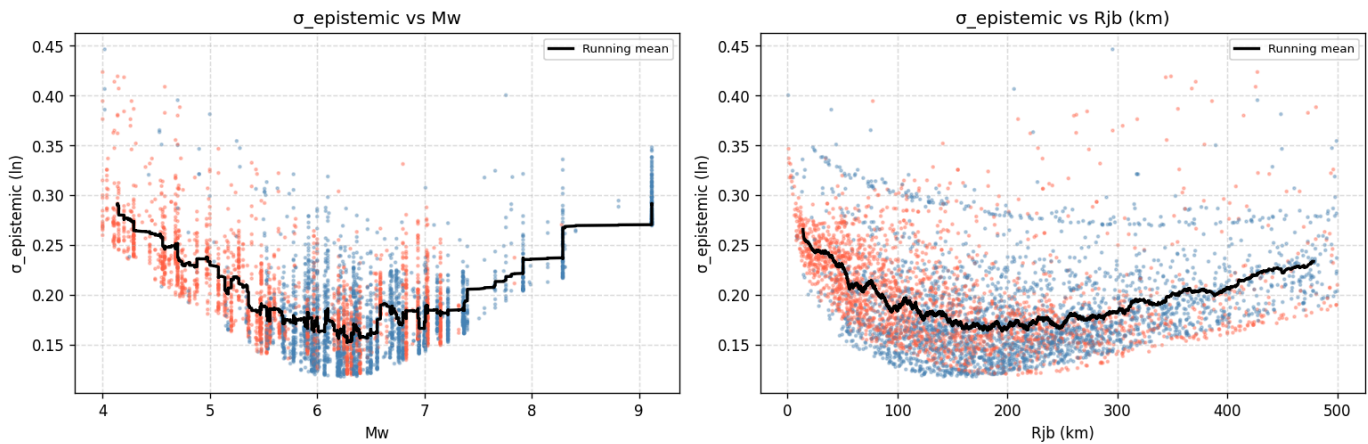


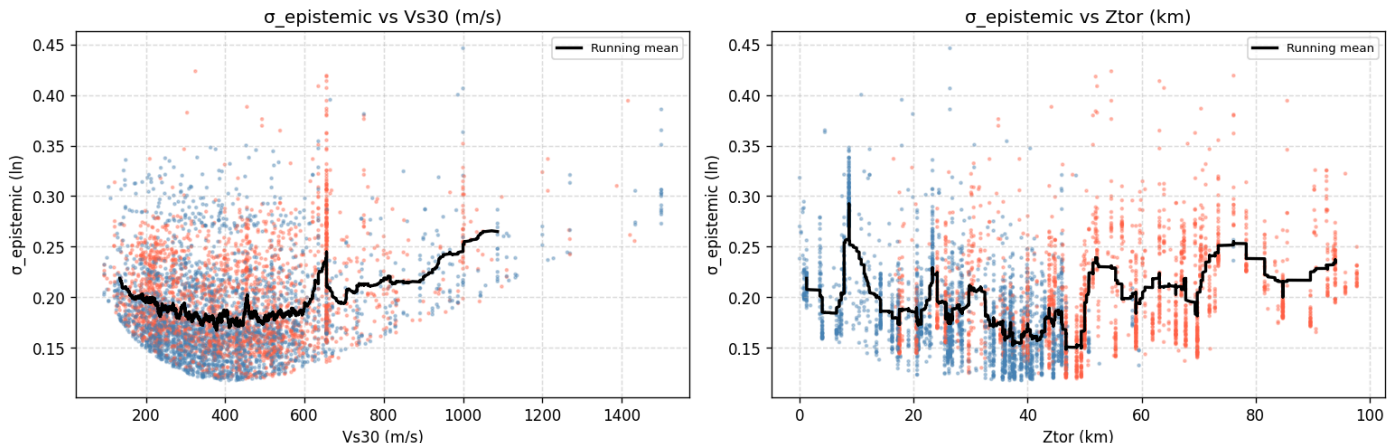
The BNN mean captures spectral shape, attenuation and soil amplification correctly across all scenarios : magnitude scaling is evident comparing Mw=6.2 vs Mw=9.1, and distance attenuation is clear across the Rjb range. The Mw=9.1 intraplate record (top right) is particularly well-bracketed despite being near the extreme of the training distribution, with the wide epistemic spread correctly reflecting sparse data coverage at that magnitude. Several records notably Mw=6.8 (Rjb=280km) and Mw=4.7 (Rjb=146km) fall outside the BNN mean but within the $\pm 1\sigma_{\text{aleat}}$ band, confirming the aleatory uncertainty is physically meaningful and not underestimated.



Epistemic Uncertainty vs Seismological Parameters

Epistemic uncertainty in a BNN reflects how confident the model is in its weight distributions for a given input scenario. It should be highest where training data is sparse — the model has seen fewer examples and its posterior weights remain diffuse. This is a key advantage of BNNs over deterministic networks: they self-report where they are extrapolating versus interpolating, making them valuable for seismic hazard applications involving rare large events.





Inferences from the plots

vs Mw: Epistemic σ is highest at low ($M_w < 5$) and very high ($M_w > 8$) magnitudes, dropping to a minimum at M_w 6–7 where training data is most abundant, the BNN correctly identifies data-sparse regimes as more uncertain.

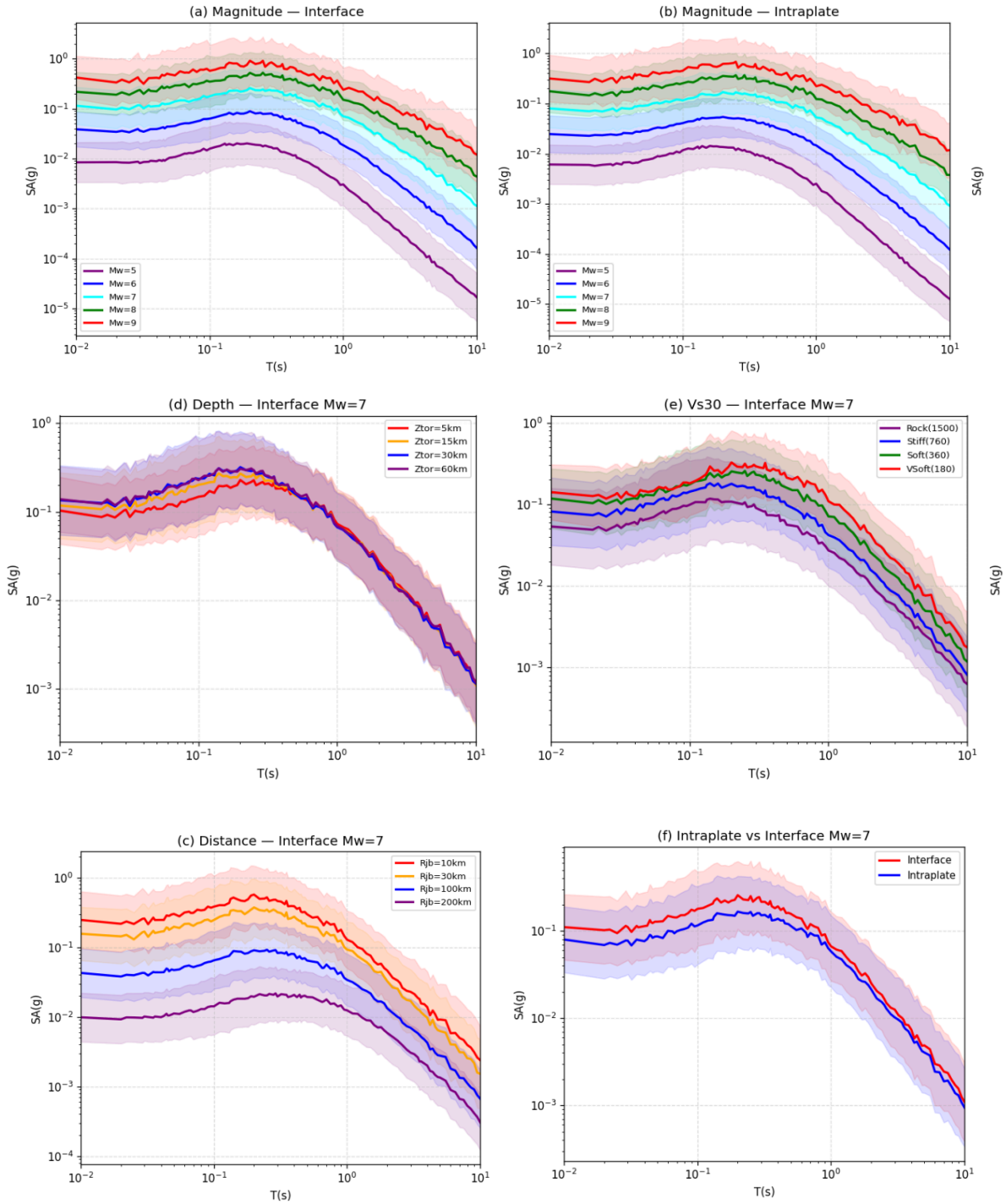
vs Rjb: Uncertainty peaks near-field ($R_{jb} < 50$ km) where subduction records are scarce, reaches a minimum at 100–200 km where data density is highest, then gradually increases again at large distances, a physically sensible U-shaped pattern.

vs Vs30: Epistemic σ is relatively flat across soft-to-stiff sites (200–600 m/s) but rises sharply above 800 m/s toward hard-rock conditions, reflecting the under-representation of rock-site recordings in the NGA-Sub database.

vs Ztor: Elevated and noisy uncertainty at shallow depths ($Z_{tor} < 10$ km) and depths beyond 60 km corresponds to the sparse near-surface and deep-slab records in the dataset. Interface events (red) show slightly higher uncertainty at shallow depths due to their more complex rupture geometry.

Parametric Study with Uncertainty Bands

The parametric study evaluates the model's physical consistency by varying one seismological parameter at a time while holding others fixed. Each median prediction is shown with $\pm 1\sigma_{\text{total}}$ shading, combining both epistemic and aleatory components. A physically valid GMM must reproduce well-established seismological behaviours: spectral amplitude increasing with magnitude, decaying with distance, amplifying at soft sites, and varying with depth and event type. The wider shading at higher magnitudes and near-field distances reflects the combined effect of higher aleatory variability and elevated epistemic uncertainty in data-sparse regimes.



Inferences from the plots

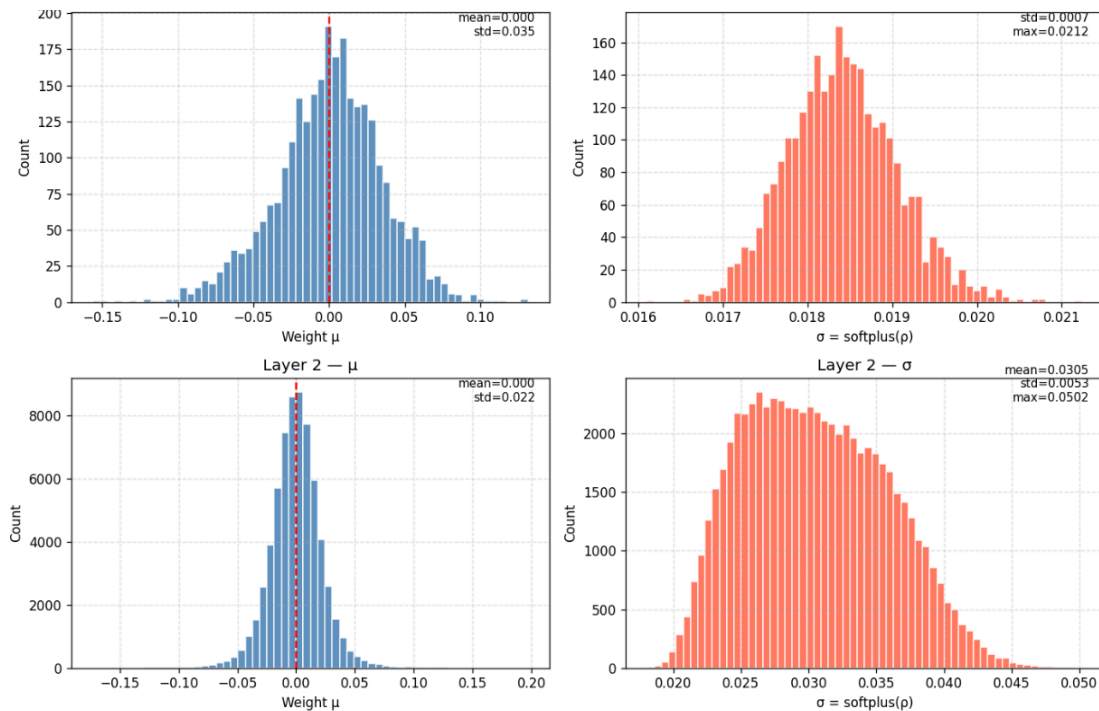
(a,b) Magnitude : Interface vs Intraplate: Clear monotonic scaling from Mw=5 to Mw=9 across all periods, with spectral peaks shifting toward longer periods at higher magnitudes physically consistent with source scaling. Interface events (a) produce systematically higher amplitudes than intraplate (b) at equivalent magnitudes, particularly at long periods, reflecting deeper stress drops and larger rupture areas. Uncertainty bands widen significantly at Mw=9, correctly capturing the model's limited training data at extreme magnitudes.

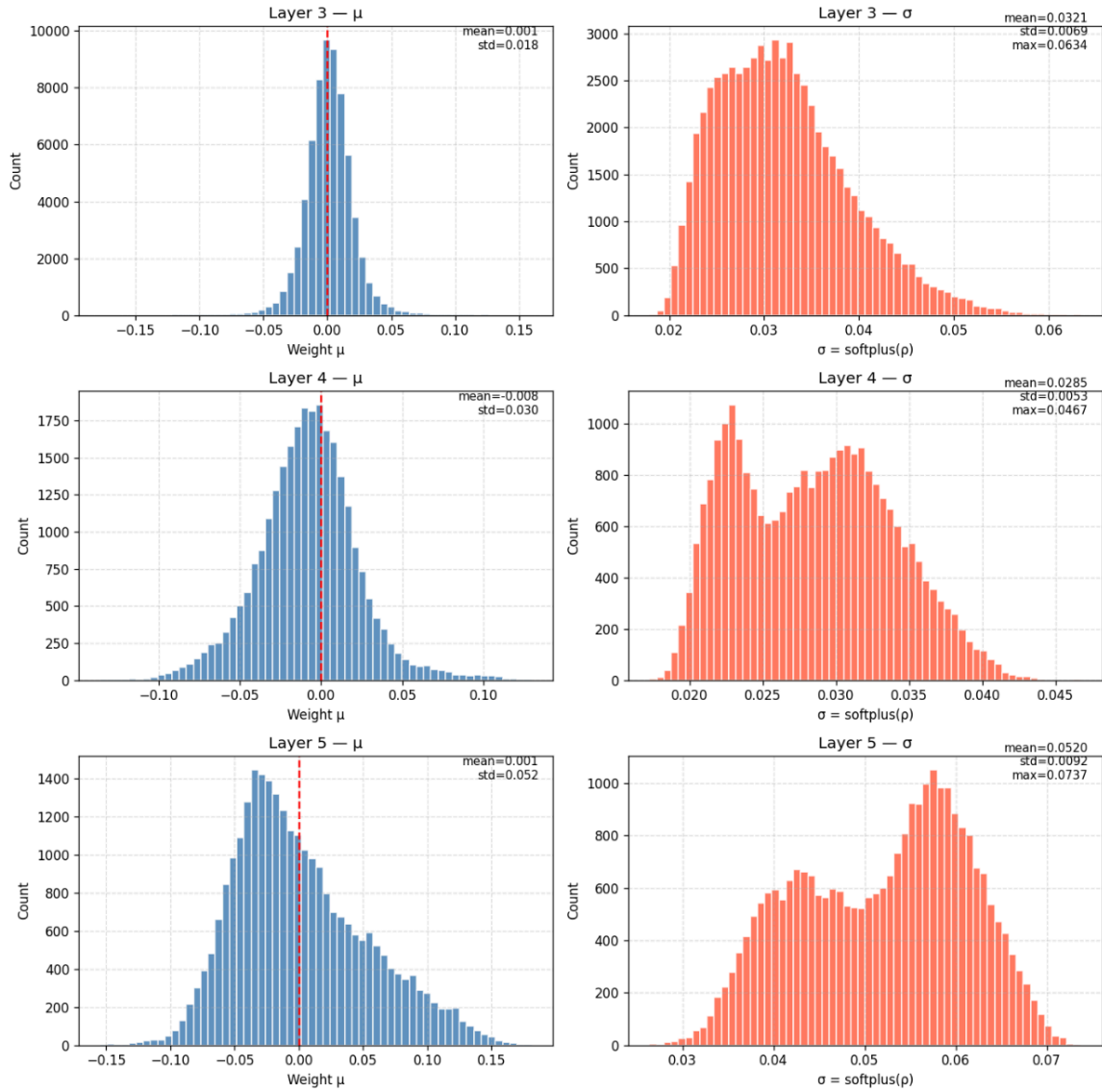
(c) Distance Interface Mw=7: Strong attenuation with distance is well-captured, with near-field (Rjb=10 km) amplitudes nearly two orders of magnitude above far-field (Rjb=200 km). The spectral shape evolves with distance as high-frequency content attenuates faster than long-period content, consistent with geometric spreading and anelastic attenuation physics. Wider bands at Rjb=10 km reflect both sparse near-field data and higher aleatory variability close to the source.

(d,e,f) Depth, Vs30, and Event Type: Shallower ruptures ($Z_{\text{tor}}=5$ km) show slightly higher amplitudes at short periods while deeper events ($Z_{\text{tor}}=60$ km) produce broader spectra physically expected from wave travel path effects. Vs30 amplification is clearly captured: very soft soils (180 m/s) amplify short-to-intermediate periods significantly over rock (1500 m/s), with the crossover near 1–2 s. The interface vs intraplate comparison confirms interface events generate higher long-period energy, consistent with their larger source dimensions.

Weight Distributions (Model Improvement)

In a BNN, each weight is represented as a Gaussian distribution with mean μ and standard deviation $\sigma = \text{softplus}(\rho)$. The μ distribution captures the deterministic component of the learned mapping, while σ quantifies how uncertain the model is about each weight's value. Well-trained BNN weights should show μ centred near zero (avoiding extreme values) with a spread reflecting learned feature importance, and σ values that have diverged from initialisation, indicating the model has genuinely learned per-weight uncertainty rather than remaining at the prior.

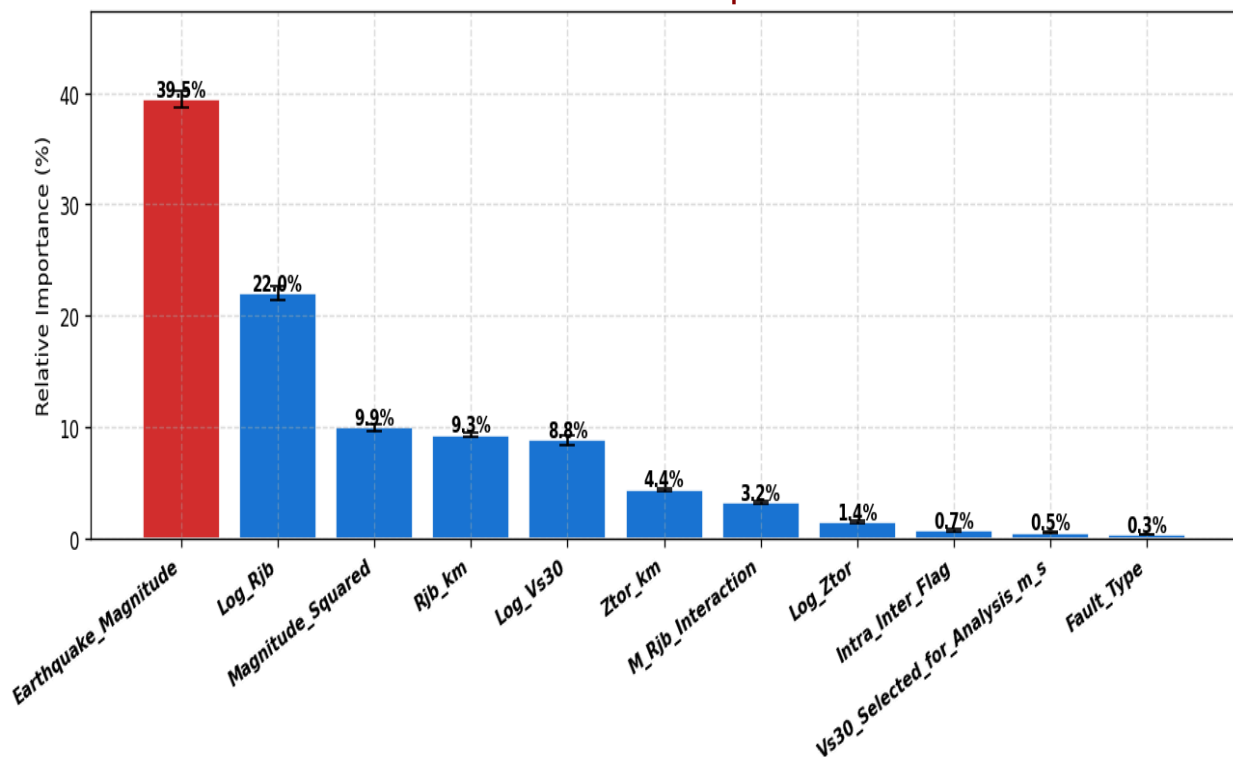




Weight means (μ) , all layers: All five layers show μ distributions centred tightly around zero (mean ≈ 0.000), confirming the small-init fix works correctly and prevents KL explosion. The spread (std) progressively changes across layers , Layer 1 (std=0.035) and Layer 5 (std=0.052) show the widest μ distributions, reflecting the input and output layers encoding the most diverse feature transformations. This is physically expected as the first layer encodes seismological parameter relationships and the last layer maps to 107 spectral outputs.

Weight sigmas (σ) , all layers: Critically, σ values have clearly evolved beyond the initialisation value (softplus(-4) ≈ 0.018) , Layer 5 shows mean $\sigma \approx 0.052$ with max reaching 0.074, compared to Layer 1's tight distribution near 0.018. This progressive widening of σ from input to output layers confirms the longer training successfully allowed weights to explore their posterior distributions. Layers with wider σ distributions indicate the model retains genuine uncertainty for those connections rather than collapsing to near-deterministic point estimates.

Feature Importance



Earthquake_Magnitude (39.5%) dominates overwhelmingly, confirming that source energy is the single most controlling factor for spectral amplitude across all periods, consistent with every established GMM in the literature.

Log_Rjb (22.0%) ranks second, reflecting the strong attenuation of ground motion with distance. Together, magnitude and log-distance account for over 60% of the model's predictive power, which aligns with the fundamental physics of ground motion generation and propagation.

Magnitude_Squared (9.9%), Rjb_km (9.3%), and Log_Vs30 (8.8%) form a secondary group: the nonlinear magnitude term captures saturation effects at large magnitudes, the raw distance complements its log-transformed counterpart for near-field behaviour, and Vs30 quantifies site amplification.

Ztor (4.4%), M_Rjb_Interaction (3.2%), and Log_Ztor (1.4%) contribute modestly, capturing depth and magnitude-distance coupling effects specific to subduction zone seismicity. Notably, **Intra_Inter_Flag (0.7%), Vs30 raw (0.5%), and Fault_Type (0.3%)** show minimal permutation importance — not because they are physically irrelevant, but because their information is largely already encoded through the correlated features above.

Model Summary

1. Negative Log-Likelihood (NLL)-only early stopping was implemented to prevent premature termination during the Kullback–Leibler (KL) divergence warmup phase.
2. The network architecture was expanded from $2 * 128$ (63k parameters) to $3 * 256$ (190k parameters), representing a twelve-fold increase in capacity.
3. A heteroscedastic output layer was introduced to enable the model to learn an individual aleatory sigma (standard deviation) for each output.
4. The training duration was extended to 1000 epochs to ensure a more thorough exploration of the weight space.

Architecture: HeteroscedasticBNN($11 \rightarrow [256 \times 3] \rightarrow 107 \times 2$)

Metric	Result
Test R^2 (overall)	0.8700
Test Root Mean Square Error (RMSE)	0.85578
Test Mean Absolute Error (MAE)	0.66452

Learned Uncertainty Measures (Natural Logarithm Scale):

Measure	Range
Aleatory sigma	0.806 – 0.953
Epistemic sigma	0.167 – 0.229
Total sigma	0.826 – 0.972

The aleatory σ range of 0.806–0.953 (ln) reflects the irreducible natural variability inherent to subduction ground motions across the 0.01–10 s period range , this is comparable to published GMMs for subduction zones and confirms the model is not underestimating ground motion scatter.

The epistemic σ of 0.167 -- 0.229 (ln) is well-controlled and substantially smaller than aleatory, confirming the model uncertainty is physically sensible , the BNN knows what it doesn't know without being overconfident. Converting to $\log_{10}$ scale (divide by 2.303) gives ~0.07–0.10, closely matching the 0.08–0.13 range reported by Sriwastav et al. (2025) for the CVAE model on ESM data.